RV College of Engineering®
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1RV22CS067

Department of CSE

Date:	**Test - 2**	**Max. Marks:60**
Semester:6th	**UG**	**Duration: 2 Hrs**
Course Title: Cryptography & Network Security		**Course Code:CS365TDF**

	Part- A	**M**	**BT**	**CO**
1.	List the criteria defined by NIST for AES.	2	1	1
2.	List the parameters (block size, key size, and the number of rounds) for the three AES versions.	2	1	3
3.	Define ECB and list its advantages and disadvantages.	2	2	2
4.	Divide the five modes of operation into two groups: those that need padding and those that do not.	2	1	1
5.	Distinguish between an MDC and a MAC.	2	2	2
	Part - B			
1.a	AES has a larger block size than DES (128 versus 64). Is this an advantage or disadvantage? Explain.	5	2	2
1.b	Write an algorithm for the InvSubBytes transformation.	5	2	1
2. a	Perform encryption and decryption using the RSA algorithm with p=7,q=11,e=17,M=8	5	3	3
2.b	Show why CFB mode creates a nonsynchronous stream cipher, but OFB mode creates a synchronous one.	5	2	4
3.a	In ECB mode, bit 17 in ciphertext block 8 is corrupted during transmission. Find the possible corrupted bits in the plaintext.	3	2	3
3.b	Prove that the plaintext used by Alice is recovered by Bob in CTR mode. Show your work with detail algorithm.	7	3	4
4.	The Rabin cryptosystem is not deterministic: Decryption creates four equally probable plaintexts. Prove the statement by taking example where p=23, q=7 and plain text is 24.	10	3	4
5.a	Mention the following in detail for Elgamal crypto system 1. Key generation algorithm 2. Encryption Algorithm 3. Decryption Algorithm	6	2	3
5.b	Write an algorithm in pseudocode for HMAC.	4	2	3